COMPARATIVE EVALUATION OF MACHINE LEARNING AND DEEP LEARNING MODELS FOR WEATHER PREDICTION IN NIGERIA USING SPATIAL ANALYSIS

OLADOYIN IDRIS ATOLAGBE; & TAYO P. OGUNDUNMADE
Department of Statistics, University of Ibadan, Ibadan, Nigeria
Corresponding Author: idrisatolagbe92@gmail.com
DOI: https://doi.org/10.70382/caijmasr.v13i9.056

ABSTRACT

Accurate prediction of meteorological variables is essential for agricultural planning, disaster risk reduction, water resource management, and climate adaptation in Nigeria, where substantial climatic variability exists across diverse ecological zones. This study presents a comparative evaluation of machine learning, deep learning, and spatial analysis techniques for predicting key meteorological variables, namely rainfall, temperature, and relative humidity, using a nationwide daily dataset comprising 270,285 observations collected across the 36 states of Nigeria between 2005 and 2024. Following comprehensive data pre-processing, spatial feature engineering was performed by incorporating geographic coordinates and spatial lag variables. Global Moran's I statistics confirmed significant positive spatial autocorrelation for rainfall, temperature, and humidity, providing empirical justification for integrating spatial information into the predictive models. Five regression models: Random Forest, Extreme Gradient Boosting (XGBoost), Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (Bi-LSTM), and Transformer were developed and evaluated using a chronological train-test split. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), the coefficient of determination (R^2), confidence intervals, and statistical comparison of prediction errors using the Diebold-Mariano test. An ablation study was further conducted to quantify the contribution of spatial features. Results demonstrated that the Transformer model achieved the best predictive performance (MAE = 1.87 mm, RMSE = 3.98 mm, $R^2 = 0.78$), representing approximately a 19% improvement in RMSE over the best-performing machine learning model (XGBoost). The ablation study showed that incorporating spatial features reduced RMSE by between 3.8% and 8.4% across all models, with the Transformer exhibiting the greatest improvement. The findings demonstrate that combining deep learning with spatial analysis substantially enhances weather prediction across Nigeria and provides a robust framework for supporting climate resilience, agricultural decision-making, and operational weather forecasting.

Keywords: Nigerian Weather Prediction; Spatiotemporal Forecasting; Transformer; LSTM; Spatial Autocorrelation; Moran's I; Machine Learning; Deep Learning.

INTRODUCTION

Nigeria's climate system is shaped by diverse geographic, hydrological, and topographic factors, producing highly heterogeneous weather patterns across its six geopolitical zones. The country spans ecological belts ranging from the humid coastal rainforests of the south to the semi-arid Sahel in the north, each characterised by distinct rainfall regimes, temperature profiles, and humidity cycles. Increasing climate volatility manifesting as erratic rainfall, prolonged dry spells, riverine flooding, and intensifying heatwaves compounds this natural variability and disproportionately affects the agricultural sector, which employs approximately 70% of the Nigerian population

(FAO, 2021). For smallholder farmers, who constitute the majority of this workforce, reliable weather forecasts are essential for crop planning, irrigation scheduling, pest management, and disaster risk reduction.

This study focuses on forecasting three key meteorological variables: daily rainfall, air temperature (maximum and minimum), and relative humidity which collectively influence agricultural productivity, water resource management, flood risk assessment, and climate resilience in Nigeria. Rainfall determines crop water availability and flood occurrence, temperature regulates evapotranspiration and crop development, while humidity influences atmospheric moisture content and precipitation processes. These variables therefore provide an appropriate basis for evaluating the predictive capability of machine learning, deep learning, and spatial analysis models.

Conventional Numerical Weather Prediction (NWP) models face well-documented limitations in the Nigerian context. The country's sparse meteorological station network particularly in the north combined with coarse-resolution global reanalysis datasets, limits the ability of NWP models to resolve localised weather anomalies and regional microclimates (Olaniyan et al., 2022). Machine learning (ML) and deep learning (DL) offer a data-driven alternative. Ensemble methods such as Random Forest and XGBoost demonstrate competence in tabular meteorological forecasting, while recurrent architectures including LSTM and Bi-LSTM, and attention-based Transformer models excel at modelling the sequential dependencies inherent in climate time-series (Rasp et al., 2020; Gao et al., 2022).

Despite this progress, few studies have rigorously compared ML and DL approaches on spatially explicit, multi-station Nigerian data while simultaneously accounting for spatial autocorrelation across all major meteorological variables. Rasp et al. (2020) noted that Random Forest struggles with sequential climate dependencies without extensive feature engineering, and that models trained on limited locations frequently fail to generalise across Nigeria's diverse climatic zones. McGovern et al. (2019) further highlighted that Transformer architectures demand computational resources that exceed the capacity of many Nigerian research institutions. Rasp et al. (2020) emphasised that spatially-aware modelling is critical for regions exhibiting strong geographic heterogeneity. Critically, no prior study has benchmarked five modelling approaches spanning both classical ML and state-of-the-art DL on a nationwide Nigerian dataset covering all 36 states, multiple meteorological variables, and nearly two decades of daily observations, while quantifying the independent contribution of spatial features through systematic ablation.

This study addresses these gaps through a systematic, reproducible benchmark of five modelling frameworks on a 20-year, nationwide daily weather dataset, with spatial features explicitly incorporated, spatial autocorrelation quantified prior to modelling, and an ablation analysis isolating the contribution of spatial encoding. The primary contributions of this work are: (i) a comprehensive ML-versus-DL benchmark using spatially augmented features on a 36-state Nigerian dataset spanning nearly two decades; (ii) empirical confirmation of statistically significant spatial autocorrelation across three Nigerian meteorological variables using global Moran's I; (iii) an ablation study quantifying the independent performance gains attributable to spatial feature integration; and (iv) a realistic phased deployment pathway that reconciles predictive performance with Nigeria's current infrastructural realities.

Aim and Objectives of the Study

The aim of this study is to develop and evaluate advanced predictive models using multivariate weather data and geospatial features for accurate forecasting of key weather parameters namely, rainfall, temperature, and humidity across Nigeria. The main objectives are to:

i. develop ML models like Random Forest (RF) and XGBoost (XGB) for weather prediction performance on multivariate datasets.
ii. implement DL models like Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), and Transformer to capture temporal dependencies and spatial heterogeneity in weather patterns.

iii. compare the performance of the models using evaluation metrics such as: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2) to assess the strengths and weaknesses of machine learning and deep learning models under varying weather conditions.

LITERATURE REVIEW

Accurate weather prediction is essential for agriculture, water resources management, disaster risk reduction and urban planning in Nigeria. Traditional Numerical Weather Prediction (NWP) systems and statistical methods have formed the backbone of operational forecasting for decades, but they face limitations in computational cost, model parameterization and the representation of local-scale processes in data-sparse regions like many parts of Nigeria. In recent years, Machine Learning (ML) and Deep Learning (DL) techniques often combined with geospatial analysis and satellite-derived datasets have been proposed as complementary or alternative approaches for local and regional forecasting. These methods offer the ability to learn complex nonlinear relationships directly from data and to fuse heterogeneous data sources such as rainfall, temperature, humidity, and vegetation indices (Reichstein et al., 2019).

Traditionally, weather forecasting is grounded in Numerical Weather Prediction which uses the physical laws of the atmosphere governed by partial differential equations describing thermodynamics, fluid motion, and mass conservation (Bauer et al., 2015). These models, such as the Global Forecast System (GFS) or the European Centre for Medium-Range Weather Forecasts (ECMWF) system, simulate future atmospheric conditions by integrating these equations forward in time (Kalnay, 2003). Although powerful, these models require immense computational resources, high-quality initial conditions, and extensive observational networks (Rasp et al., 2020).

By contrast, deep learning has revolutionized time-series forecasting in meteorology. Architectures such as Long Short-Term Memory (LSTM) networks and their Bidirectional LSTM (Bi-LSTM) variants effectively capture long-term dependencies and sequential temporal patterns (Shi et al., 2015). These methods have demonstrated superior performance in rainfall prediction, wind speed estimation, and temperature forecasting compared to traditional statistical models (Rasp et al., 2020). However, limited research has focused specifically on Nigeria. Region-specific challenges include low station density, complex orography (e.g., the Jos Plateau), and strong seasonal influences of the West African monsoon, which introduce variability in rainfall and temperature patterns (Arowolo et al., 2017; Akinsanola et al., 2017). Integrating spatial analysis and remote sensing data such as MODIS and CHIRPS with deep learning models has been identified as a promising approach to overcome data scarcity and improve localized forecasting (Reichstein et al., 2019).

Empirical Review

A review of the literature shows that Machine Learning models (Random Forest, XGBoost) and Deep Learning models (LSTM, Bi-LSTM, Transformer) are widely used in spatial research. A number of articles are specifically looked at with the intention of employing deep learning for spatial analysis.

McGovern et al. (2019) identified computational and infrastructural barriers of ML and DL models, especially architectures like transformers and ConvLSTMs, demand substantial computational resources for training and deployment. Nigeria's research institutions and meteorological agencies often lack the necessary high-performance computing infrastructure. Furthermore, limited internet penetration in rural areas and unreliable electricity supply exacerbate the challenge of implementing real-time, data-intensive weather prediction systems. Olaniyan et al. (2022) stated that the Nigerian Meteorological Agency (NiMet) operates observational stations across the country, but their distribution is uneven, with dense coverage in southern regions and sparse coverage in the north. This imbalance leads to spatial data gaps that hinder model generalization. Arowolo et al. (2017) highlighted that missing values, inconsistent measurement standards, and inadequate digitization of historical records compromise data

quality, and satellite data, while useful, often suffer from coarse spatial resolution and biases specific to West African climates.

An approach proposed by Rasp et al. (2020) stated that several methodological issues constrain the use of ML/DL in Nigerian weather forecasting temporal modelling, algorithms like Random Forest struggle with sequential dependencies unless significant feature engineering is performed. Spatial modelling most studies ignore cross-regional weather dependencies, leading to localized models that fail to generalize across Nigeria's climatic zones, and generalizability of ML/DL models trained on data from limited locations often overfit and perform poorly when applied elsewhere. Moreover, Nigeria's institutional frameworks for weather monitoring and climate services face challenges such as inadequate funding, bureaucratic bottlenecks, and limited collaboration between academia, government, and industry. The lack of open-access meteorological datasets restricts knowledge sharing and hinders reproducibility of research. Furthermore, there is limited policy integration of advanced ML/DL forecasting outputs into agricultural planning, flood management, or disaster preparedness.

Gao et al. (2022) discovered Earthformer, a Transformer-based architecture that employs space-time attention to forecast precipitation and ENSO dynamics. The model achieved state-of-the-art performance on real-world benchmarks, highlighting the capacity of attention-based spatiotemporal models for Earth system prediction This directly supports the integration of Transformer architectures into weather and climate modeling. Robin et al. (2022) evaluated ConvLSTM models to forecast fine-resolution vegetation greenness across Africa by combining Sentinel-2 data with ERA5 reanalysis and topographical features, their work underscored the importance of integrating spatial and temporal information for environmental prediction, while demonstrating ConvLSTM's ability to model spatially resolved, climate-sensitive variables.

Nyasulu et al. (2023) focused on rainfall prediction, comparing Gradient Boosting, CatBoost, Random Forest, and Ridge Regression models for daily rainfall forecasting in West Africa. Gradient Boosting achieved the highest performance ($R^2 \approx 0.69$) with the lowest RMSE, setting a benchmark for ML-only approaches in the region. Rasp et al. (2020) explored the interpretable ML methods, using EOF-driven feature reduction for seasonal precipitation forecasting in East Africa, their approach, which delivered comparable or superior results to ECMWF seasonal forecasts in certain periods, illustrated the operational feasibility of interpretable ML models driven by spatial indices. Deep learning has also been successfully applied to short- and medium-range rainfall forecasts. U-Net convolutional networks trained on ERA5 and GPM-IMERG datasets to forecast 24-hour rainfall at 12 h and 30 h lead times, their model matched or exceeded ECMWF benchmarks, with added benefits of interpretability analyses, demonstrating the effectiveness of spatially aware CNN-based architectures for West African rainfall forecasting (Rasp et al., 2020).

Ravuri et al. (2021) incorporated tropical wave features into gamma regression and CNN models trained on GPM-IMERG data, generating calibrated probabilistic forecasts that outperformed ECMWF baselines. This highlights the value of incorporating domain-specific spatial predictors, such as tropical waves, in rainfall prediction frameworks-an approach with strong relevance for Nigerian monsoonal rainfall.

El Hachimi et al. (2024) identified the robustness of deep learning in semi-arid climates compared LSTM with classical statistical regression models for short-term forecasting in Morocco. LSTM consistently outperformed regression models for multivariate precipitation and temperature, reinforcing DL's superiority in semi-arid contexts like northern Nigeria. In contrast, Samson et al. (2024) compared Random Forest and LSTM on monthly rainfall data between 1980–2022 from six cities in southwestern Nigeria, finding Random Forest to be superior across all accuracy metrics, this result underscores the role of regional context and data availability in determining whether ML or DL methods are most effective.

Gao et al. (2022) discovered hybrid approaches are also emerging as powerful alternatives. Developed finres_S2S, a hybrid cluster-grid point ML framework for seasonal precipitation and temperature forecasting. The model

preserved spatial resolution while reducing computational cost, outperforming ECMWF forecasts up to six-month lead times. Nyasulu et al. (2023) compared Gradient Boosting, CatBoost, Random Forest, and Ridge Regression for daily rainfall prediction in the Sahel region; Gradient Boosting achieved the strongest reported performance among the tested regressors. applied DL techniques to weather prediction in Africa. For instance, his results show how LSTM models for rainfall forecasting in Ghana, demonstrating improved accuracy compared to ARIMA and support vector regression. However, these studies share common limitations: Most focus on single-country or single-region datasets, limiting generalizability. There is a lack of comparative evaluation between DL and ML models in African weather contexts. Demissie et al., (2024) incorporated spatial variables in a meaningful way, despite clear regional climatic disparities. Furthermore, there is limited exploration of transformer-based models for weather prediction in sub-Saharan Africa. Given their success in other domains, this represents a significant research opportunity. A key methodological gap lies in the reliance on strictly temporal models that treat weather data as independent time series at single stations or locations. This neglects the inherent spatial dependence across meteorological variables. For instance, rainfall in northern Nigeria is strongly influenced by Sahelian weather systems, while southern Nigeria is driven by coastal and Atlantic Ocean dynamics. Ignoring these spatial dependencies leads to models that may perform well locally but fail to generalize when deployed across diverse climatic regions (Nyasulu et al., 2023).

Robin et al. (2022) employed relatively simple evaluation frameworks, such as RMSE and MAE, without examining model calibration, uncertainty quantification, or spatial transferability. These limitations make it difficult to draw reliable conclusions about the true utility of DL methods for operational forecasting in African contexts. Another underexplored area is the integration of multivariate inputs that reflect both atmospheric conditions and broader climatic drivers. While temperature, rainfall, and humidity are often included, few studies incorporate exogenous variables such as sea surface temperature anomalies, vegetation indices, or topographic features, all of which exert strong influence on weather dynamics in Nigeria. The absence of such multivariate and spatially aware approaches restricts the ability of existing models to capture complex interactions, such as the influence of the Inter-Tropical Discontinuity (ITD) on rainfall variability across Nigeria. Vaswani et al. (2017) mentioned how transformers have revolutionized sequence modeling in natural language processing and achieved state-of-the-art results in time-series forecasting tasks, their application in African weather prediction remains virtually unexplored. The attention mechanism in transformers allows the model to capture long-range temporal dependencies without the vanishing gradient problems common in recurrent networks. This makes them especially well-suited for rainfall forecasting in Nigeria, where weather patterns are strongly seasonal and influenced by long-term climatic drivers such as the El Niño–Southern Oscillation (ENSO), a systematic evaluation of transformers against more established DL methods like LSTM, Bi-LSTM, and ConvLSTM is therefore warranted to assess their suitability for African meteorological data.

Arowolo et al. (2017) showed how integrating observational station data with auxiliary spatial information (e.g., elevation, land cover, distance to coastlines) and temporal climate indices, the dataset allows for more realistic representation of Nigeria's diverse climatic zones. This approach ensures that models are evaluated not just on their ability to forecast at specific locations but also on their robustness across spatially heterogeneous regions. In addition, this study implements a comparative evaluation of both classical machine learning models (e.g., Random Forest, XGBoost) and state-of-the-art deep learning architectures (LSTM, Bi-LSTM, ConvLSTM, and Transformers). By benchmarking these models under consistent experimental conditions, the research provides a clearer understanding of the relative strengths and weaknesses of ML versus DL approaches in African weather forecasting. Furthermore, uncertainty quantification is incorporated through ensemble methods and probabilistic scoring metrics, ensuring that forecasts are not only accurate but also reliable.

Endalie et al. (2022) mentioned how modeling both temporal and spatial dependencies contribute to more equitable and geographically robust forecasting systems. Ultimately, this research represents attempts to systematically evaluate and integrate deep learning and spatial methods for weather prediction in Nigeria. The integration of spatial analysis techniques, including kriging and spatial cross-validation, further strengthens this study by explicitly evaluating the transferability of models across regions with differing climatic regimes. This spatially explicit evaluation framework is critical in Nigeria, where observational station density is uneven and some regions such as the northeast are underrepresented in data collection (Samson & Aweda, 2024).

Research Gaps

Despite the growing interest in machine learning (ML) and deep learning (DL) for meteorological applications worldwide, research within the Nigerian context remains relatively underdeveloped. Several gaps hinder the full exploitation of these advanced techniques for accurate weather forecasting across the country's diverse climatic zones.

i. Underrepresentation of Spatiotemporal Modeling Approaches: Most Nigerian weather studies using ML/DL adopt time-series forecasting at individual stations without adequately incorporating spatial heterogeneity (Demissie et al., 2020).

ii. Lack of Comparative Evaluations Between Classical ML and DL Models: The majority of Nigerian studies employing ML/DL for weather prediction evaluate models in isolation without systematic comparisons against both statistical baselines (e.g., ARIMA, SARIMA) and other ML/DL architectures (e.g., Random Forests vs. LSTMs vs. Transformers). This lack of benchmarking makes it difficult to determine the true added value of DL in the Nigerian meteorological context. In many cases, simpler models could perform competitively given limited data, but this is rarely quantified. A structured comparative evaluation framework remains a major gap (Robin et al., 2022).

iii. Insufficient Integration of Exogenous and Multivariate Predictors: Most existing Nigerian ML/DL weather prediction studies rely primarily on core meteorological parameters such as temperature, rainfall, and humidity. Yet, Nigeria's climate is strongly influenced by large-scale atmospheric and oceanic drivers such as sea surface temperature anomalies, vegetation indices, land cover changes, and topography. These exogenous variables are rarely incorporated into predictive frameworks, despite their potential to improve forecast accuracy. Similarly, socioeconomic and environmental variables relevant to hydrology and agriculture (e.g., soil moisture, river discharge) are seldom considered, limiting the practical utility of forecasts for decision-making (Nyasulu et al., 2023).

iv. Minimal Exploration of Transformer-Based Architectures: While LSTM and Bi-LSTM models are beginning to appear in Nigerian weather research, Transformer architectures widely regarded as state-of-the-art for long-sequence modeling remain almost unexplored. Transformers' attention mechanism is especially promising for capturing seasonal and inter-annual dependencies such as monsoon onset and ENSO-linked variability. Their absence from Nigerian-focused weather prediction studies represents both a methodological and technological gap (Rasp et al., 2020).

v. Limited Work on Uncertainty Quantification and Probabilistic Forecasting: Most ML/DL studies in Nigeria produce deterministic forecasts without accounting for predictive uncertainty. However, uncertainty quantification is crucial in meteorology, where predictions inform high-stakes decisions in agriculture, flood management, and energy planning. Probabilistic frameworks, ensemble learning, Bayesian DL approaches, and metrics such as CRPS (Continuous Ranked Probability Score) are rarely applied in Nigerian contexts. As a result, forecasts may appear accurate on average but fail to convey the degree of confidence required for operational use Ravuri et al. (2021).

vi. Computational and Infrastructure Constraints: Deep learning models demand significant computational power, which is often lacking in Nigerian research institutions. Limited access to high-performance computing (HPC), GPUs, and cloud-based infrastructure has restricted the application of cutting-edge architectures such as ConvLSTMs, Transformers, and Graph Neural Networks. Consequently, many Nigerian studies opt for relatively simple models, leaving unexplored the potential of advanced DL for operational-scale weather forecasting (Rasp et al., 2020).

vii. Weak Integration with Operational Forecasting Systems: Another major gap is the lack of integration between academic ML/DL research and operational systems used by the Nigerian Meteorological Agency (NiMet). Forecasting research often remains at the proof-of-concept stage, with little translation into real-time, user-oriented platforms for agriculture, disaster management, and public safety. Bridging this gap requires collaboration between researchers, policymakers, and industry stakeholders, as well as the development of deployable models with user-friendly interfaces.

viii. Lack of Longitudinal, Multi-Region Studies: Finally, there is an overconcentration of ML/DL research on a few urban or well-instrumented regions (e.g., Lagos, Ibadan, Abuja), while other regions remain underrepresented. These geographical bias skews results and limits insights into nationwide applicability. Large-scale, multi-region studies that evaluate model performance across Nigeria's ecological zones (Sahel, savanna, rainforest, mangroves) are virtually absent.

MATERIALS AND METHODS

Dataset Description and Source

Daily weather records for all 36 Nigerian states (plus the Federal Capital Territory, Abuja) from January 2005 to December 2024 were obtained from the Nigerian Meteorological Agency (NiMet). Data were acquired as station-level observations and subsequently aggregated to state-level daily summaries by computing spatial means over all stations within each state boundary. Key variables include maximum temperature (°C), minimum temperature (°C), rainfall (mm), relative humidity (%), and geographic coordinates (latitude, longitude) of state centroids. The complete dataset comprises 270,285 daily state-level records, providing a robust and temporally deep foundation for model training and evaluation across Nigeria's six geopolitical zones and diverse climatic belts.

Data Preprocessing

Data Cleaning

Raw station data were screened for duplicate records, inconsistent measurement units, and physically implausible values, such as negative rainfall or temperatures exceeding established climatological bounds for Nigeria (World Meteorological Organization, 2018). Outliers were identified via interquartile range (IQR) analysis (flagging values beyond Q1 - 3×IQR and Q3 + 3×IQR) and World Meteorological Organization threshold guidelines. Duplicate observations were removed prior to all further processing steps.

Missing Value Imputation

An initial data quality audit revealed an overall missingness rate of 6.2% across all variables, with the highest rates observed in northern states (Borno: 11.4%, Yobe: 9.8%) due to limited station density. Missing values arising from sensor outages and incomplete station coverage were addressed using a two-stage hybrid imputation strategy. Temporal gaps of three or fewer consecutive days were filled via linear interpolation to preserve temporal continuity. Longer gaps were imputed using Inverse Distance Weighting (IDW) spatial interpolation from neighbouring state observations, following the method of Shepard (1968), with a power parameter of 2 and a neighbourhood radius encompassing the three nearest states. Variables exhibiting more than 20% missingness were

excluded to prevent the introduction of systematic bias into the modelling pipeline; no variable exceeded this threshold in the final dataset.

Train-Test Split and Normalisation

Data were partitioned chronologically: records from January 2005 to December 2020 (80%; n = 213,163) constituted the training set, while records from January 2021 to December 2024 (20%; n = 57,122) constituted the held-out test set. This chronological split prevents data leakage and ensures that all reported performance metrics reflect genuine out-of-sample generalisation. All continuous predictors were standardised using z-score normalisation ($X^* = (X - \mu) / \sigma$), computed exclusively from training-set statistics and applied identically to the test set.

Spatial Feature Engineering

To explicitly account for geographic heterogeneity across Nigerian states, two categories of spatial features were incorporated into all models. First, each observation was augmented with the latitude and longitude of the corresponding state centroid, enabling models to learn large-scale spatial gradients. Second, spatial lag variables were constructed using a queen-contiguity spatial weights matrix W (row-standardised), computing for each variable X at location i:

$$WX_i = \sum_j W_{ij} X_j$$

where W_{ij} represents the row-standardised spatial adjacency weight between states i and j. These spatial lag features capture inter-state climatic influence arising from shared atmospheric circulation patterns. For ML models, spatial features were concatenated with meteorological predictors; for DL models, they were appended to temporal input sequences to facilitate joint spatiotemporal learning. The independent contribution of these spatial features was subsequently quantified through ablation analysis (Section 2.7).

Model Architectures

Random Forest

Random Forest is an ensemble method that aggregates predictions from multiple independently trained decision trees via majority voting (classification) or averaging (regression). It's called a "forest" because it aggregates the predictions of multiple trees to improve accuracy and control overfitting. Hyperparameters were tuned via random search with 5-fold cross-validation. The final configuration comprised 300 estimators, a maximum depth of 20, and a minimum of 3 samples per leaf.

Classification ($majority\ vote$): $\hat{y} = mode = h_1(x), h_2(x), \ldots, h_n(x)$

Regression ($average\ prediction$): $\hat{y} = \frac{1}{n}\sum_{i=1}^{n} h_i(x)$

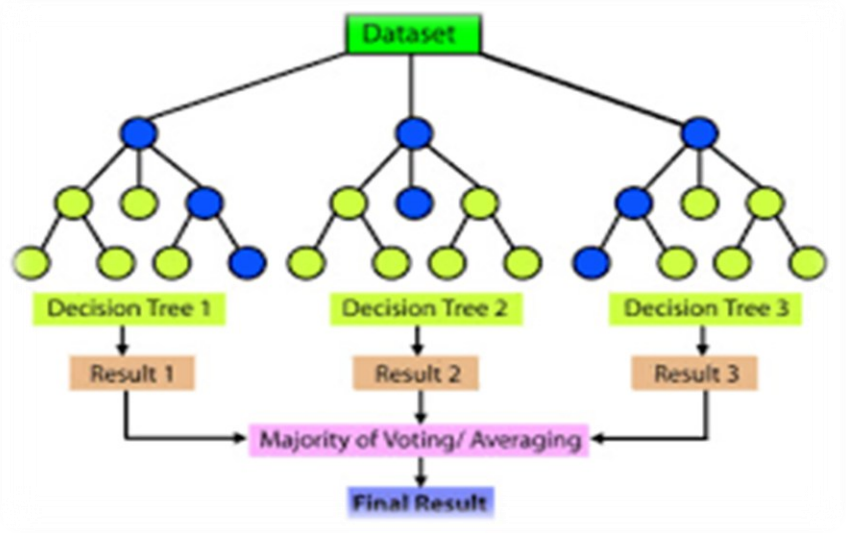


Figure 1: Random Forest Architecture (Source: Chowdhury, 2023)

XGBoost

XGBoost implements regularised gradient boosting using second-order Taylor approximations of the loss function. At each iteration t, the objective $L(t) = \sum_{i=1}^{n} l\left(y_i, y_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t)$ is minimised, where Ω penalises model complexity through tree depth and leaf weight norms. The final configuration used a learning rate of 0.05, maximum depth of 6, and a subsample ratio of 0.8.

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{i=1}^{T} wj^2 \; \Omega(f): Regularization\ term$$

$Gradient\ and\ Hessian$: $Using\ second-order\ Taylor\ approximation$:

$$L(t) \approx \sum_{i=1}^{n} = [g_i f_t(x_i) + \frac{1}{2} f_t{}^2(x_i)] + \Omega(f_t)$$

$Where$:

l: $differentiable\ convex\ loss\ function$ ($squared\ error$)

f_t: $the\ tree\ added\ at\ iteration\ t$

T: $number\ of\ leaves$

w_j: $weight\ of\ leaf\ j$

γ, λ: $regularization\ parameters$

$g_i: \frac{\partial}{\partial \hat{y}^{(t-1)}}(yi, \hat{y}^{t-1})$ *is the Gradient*

$h_i: \frac{\partial^2}{\partial \hat{y}^{(t-1)^2}}(yi, \hat{y}^{t-1})$ *is the Hessian*

LSTM

Long Short-Term Memory networks (Hochreiter and Schmidhuber, 1997) overcome the vanishing gradient problem through gated cell memory. The forget gate $f_t = \sigma(W_n[h_{t-1}, X_t] + b_f)$ selectively discards irrelevant historical information; the input gate and candidate state jointly update cell memory; and the output gate controls

the hidden state passed to subsequent layers. A two-layer LSTM with 128 hidden units per layer was used, with dropout (rate = 0.2) applied between layers.
Mathematically this is represented as,
Forget Gate (f_t): decide what to discards
$f_t = \sigma\,(W_f \,.\, [h_{t-1}, X_t] + b_f\,)$

Input Gate (i_t): decides what new information to store
$i_t = \sigma\,(W_i \,.\, [h_{t-1}, X_t] + b_i\,)$

Output Gate (O_t): decides the next hidden states
$O_t = \sigma\,(W_O \,.\, [h_{t-1}, X_t] + b_O\,)$

Candidate Cell State: proposed $update\ to\ the\ internal\ cell\ state\ at\ a\ step\ t$
$C^{\sim} = \tanh\,(W_c \,.\, [h_{t-1}, X_t] + b_c\,)$

Update Cell State: process of updating internal cell state at each time step t.
$U_t = f_t * [C_{t-1}, i_t] \ * C^{\sim}$

Hidden State: internal representation of the input sequence at each time step t.
$h_t = O_t * tanh\,(C_t)$

$Where$:
X_t: $Input\ vector\ at\ time\ t$,
h_{t-1}: $hidden\ state\ from\ previous\ step$,
C_{t-1}: $cell\ state\ from\ previous\ step$,
C_t: *cell state*,
W_c: $weight\ matrix\ for\ candidate\ state$,
b_c: $bias\ term$,
tanh: the activation function to keep values between -1 and 1,
σ: $sigmoid\ function$,
$*$: $element\text{-}wise\ multiplicatio$n

Bidirectional LSTM (Bi-LSTM)

Bidirectional LSTM (Graves and Schmidhuber, 2005) processes input sequences in both forward and backward directions, concatenating the resulting hidden states: $h_t = [h_t\rightarrow, h_t\leftarrow]$. This architecture captures both past and future temporal context, which is beneficial for modelling seasonal climate patterns where future conditions inform interpretation of current observations. The same two-layer, 128-unit architecture and dropout configuration as LSTM were employed.

Transformer

The Transformer architecture (Vaswani et al., 2017) replaces recurrence with multi-head self-attention, enabling parallel processing of entire sequences and direct modelling of long-range dependencies. Scaled dot-product attention is computed as:

Attention (Q, K, V) = softmax ($QK^T/\sqrt{d_k}$)V,

where Q, K, V are query, key, and value matrices derived from the input. A six-layer encoder-only Transformer with eight attention heads, model dimension d_{model} = 256, and feedforward dimension 512 was implemented. Positional encodings PE (pos, 2i) = sin (pos/$10000^{2i/d_{model}}$) inject temporal order into the position-agnostic attention mechanism.

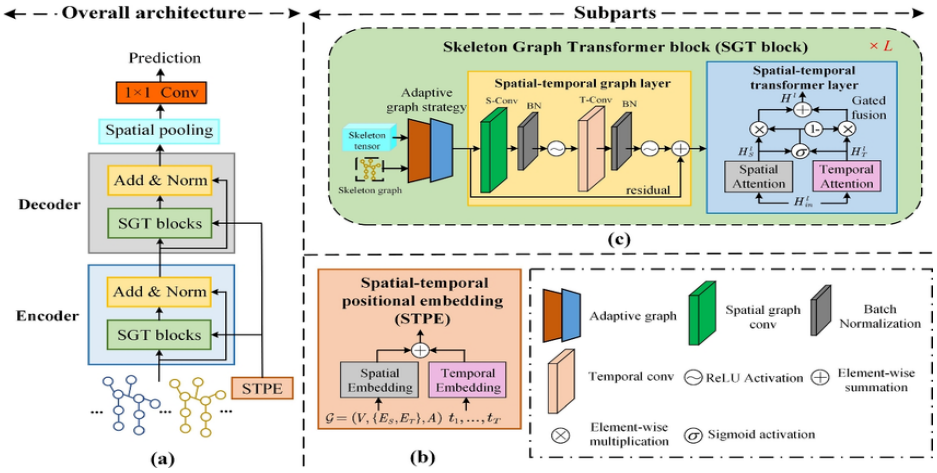


Figure 2: Architecture of transformer in spatial analysis (Source: Xiaoyan et al., 2023)

Spatial Analysis

Spatial analysis is a technique used to analyze and understand patterns, relationships, and trends in geographic data. It involves examining data with spatial components, like maps, satellite images, or GPS coordinates to extract insights and inform decisions.

Spatial Analysis Model

Spatial representation of weather data: the weather observations are indexed by spatial location $S_i = (lat_i, lon_i)$ and time t. Let $y_{i,t} \in R$ denote the observed weather variable (rainfall, temperature, humidity) at location i and time t.

Spatial Autocorrelation Modelling

Spatial autocorrelation describes the statistical independence of observations across geographic space, where nearby locations tend to exhibit similar values, violating the assumption of spatial independence and necessitating spatially explicit modeling approaches. Weather variables such as rainfall, temperature, and humidity exhibit spatial dependence due to atmospheric circulation, topography, and proximity to large water bodies. Observations recorded at nearby locations are therefore not statistically independent. To quantify this spatial dependence and justify the use of spatially-aware deep learning models, spatial autocorrelation was introduced using Moran's I statistic.

Moran's I:

This measures overall spatial dependence across the entire study area (correlation-based).

$$I_t = \frac{n}{\sum_i \sum_j W_{ij}} \cdot \frac{\sum_i \sum_j W_{ij} (y_{i,t} - \bar{y})(y_{j,t} - \bar{y})}{\sum_i (y_i - \bar{y})^2}$$

A significant I_t confirms spatial dependence in Nigerian weather pattern.

When $I_t > 0$: positive autocorrelation, $I_t < 0$: negative autocorrelation, $I_t \approx 0$: random

Where:

n is number of spatial units (stations),

I_t is the value of location i,

$\bar{y}$ = mean of variable,

W_{ij} is spatial weight between locations i and j.

$W = \sum_i \sum_j W_{ij}$

Spatial Regression Modelling

Spatial regression refers to a class of regression models that explicitly account for statistical dependence and spatial heterogeneity in geographically referenced data. It is used when the classical regression assumption of independent errors is violated due to spatial autocorrelation.

$$y = \rho W y + X\beta + WX\theta + \varepsilon$$

Geographically Weighted Regression:

$$y_i = \beta_0 (u_i, v_i) + \sum_k \beta_k (u_i, v_i) x_{ik} + \varepsilon_i$$

At location i, a one-unit increase in x_k is associated with a $\beta_k (u_i, v_i)$ change in y, holding other variables constant. Where y is the vector of dependent variables, X is the matrix of explanatory variables, W is the spatial weights matrix, ρ measures the strength of spatial dependence in the dependent variable, and θ captures spatial spill over effects of the covariates, (u_i, v_i) are the spatial coordinates, $\beta_k (u_i, v_i)$ is the local coefficient for predictor k at location i.

Computational Environment and Reproducibility

All analyses were conducted using Python 3.11.5. Machine learning models were implemented using Scikit-learn 1.5.0, XGBoost 2.1.0, while deep learning models were developed using PyTorch 2.3.0. Spatial analyses were performed using GeoPandas 0.14.4, PySAL 24.1, and NumPy 1.26. Model training was performed on a workstation equipped with: Intel Core i7 processor, 32 GB RAM. To ensure reproducibility, all random processes including parameter initialization, data shuffling, and model training were initialized using a fixed random seed of 42**.** Hyperparameter tuning was performed using identical search spaces across repeated runs, and the chronological train-test partition remained fixed throughout all experiments.

Performance Metrics

All metrics were computed separately for each target variable (rainfall, maximum temperature, minimum temperature, humidity) and averaged to yield overall model performance scores. RMSE is reported as the primary ranking metric due to its sensitivity to large prediction errors, which are particularly consequential for operational weather forecasting.

Mean Absolute Error (MAE)

Mean Absolute Error is a commonly used metric to evaluate the accuracy of a regression model. It measures the average magnitude of errors between predicted and actual values, without considering their direction, it doesn't matter if the prediction is above or below the actual value.

MAE = $\frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$

Root Mean Square Error (RMSE)

Root Mean Square Error is a standard way to measure the accuracy of a regression model. It calculates the square root of the average of the squared differences between actual and predicted values, and it penalizes large errors more than MAE. It is used widely in climate and time-series prediction tasks. RMSE tells us how far our model predictions are from actual values, on average, in the same units as the output variable (e.g., °C, mm, etc.).

$$\text{RMSE} = \sqrt{\sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)^2}{n}}$$

Coefficient of Determination (R^2 *Score*)

The coefficient of determination is a regression metric that measures how well the model's predictions explain the variability of the actual (observed) data. R^2 tells us what percentage of the variation in the dependent variable (e.g., temperature, rainfall) is explained by the model**.** It ranges from to 1, the higher the coefficient of determination, the better the model's predictions showing that the model explains a significant portion of the variance. R^2 is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(yi - \hat{y}i)^2}{\sum_{i=1}^{n}(yi - \bar{y}i)^2}$$

In addition to MAE, RMSE, and $\boldsymbol{R^2}$, statistical significance testing was performed using the Diebold-Mariano (DM) test to determine whether differences in forecast between competing models were statistically significant rather than arising from random sampling variability.

Statistical Significance Testing of Predictive Performance

To determine whether the observed differences in predictive performance among the machine learning and deep learning models were statistically significant, the Diebold-Mariano (DM) test (Diebold & Mariano, 1995) was employed. The DM test is widely used in forecasting studies to compare the predictive errors of two competing models using out-of-sample forecasts. Unlike descriptive performance measures such as MAE, RMSE, and R^2 which summarize model performance, the DM test evaluates whether the difference in prediction errors between two models is statistically significant.

The null hypothesis states that the competing models exhibit equal predictive performance, whereas the alternative hypothesis states that one model produces significantly lower prediction errors than the other.

$$H_0: \text{E}(L_1 - L_2) = 0$$

$$H_1: \text{E}(L_1 - L_2) \neq 0$$

Where L_1 and L_2 denote the loss functions (squared forecast errors) of the two competing forecasting models.

For each pairwise comparison, forecast errors were computed on the independent test dataset using squared-error loss,

$$L_i = (yi - \hat{y}i)^2,$$

where yi represents the observed rainfall value and $\hat{y}i$ denotes the corresponding model prediction. The DM statistic was then calculated from the mean loss differential across the testing period. Statistical significance was assessed at the 5% significance level (α=0.05). A p-value less than 0.05 indicates that the observed difference in predictive performance is unlikely to have occurred by chance.

IJMASR: E-ISSN 3027-2122 P-ISSN 3027-0545

RESULTS

Descriptive Statistics

Table 1 below summarises the descriptive statistics of the meteorological variables across all 270,285 records. The mean maximum temperature of 31.997°C and minimum of 23.996°C reflect Nigeria's tropical thermal regime. The full temperature range (23.40 to 41.20°C) captures the substantial contrast between cooler southern coastal zones and the hotter Sahelian north. Mean daily rainfall of 2.010 mm, averaged across all states and all seasons, is consistent with expectations for a continental-scale dataset that includes both arid northern states and high-rainfall southern belts. Humidity ranges from 63% to 100%, further reflecting this north-south climatic gradient.

Table 1. Descriptive Statistics of Meteorological Variables (2005-2024, n = 270,285)

Statistic	Temp Max (°C)	Temp Min (°C)	Rainfall (mm)	Humidity (%)	Latitude	Longitude
Count	270,285	270,285	270,285	270,285	256,481	256,481
Mean	31.997	23.996	2.010	84.989	8.608	8.844
Std Dev	2.130	2.003	2.003	2.017	2.608	3.089
Min	23.40	14.60	0.00	63.00	4.33	2.88
Median	32.00	24.00	1.40	85.00	9.4700	7.55
Max	41.20	32.50	23.70	100.00	13.58	14.51

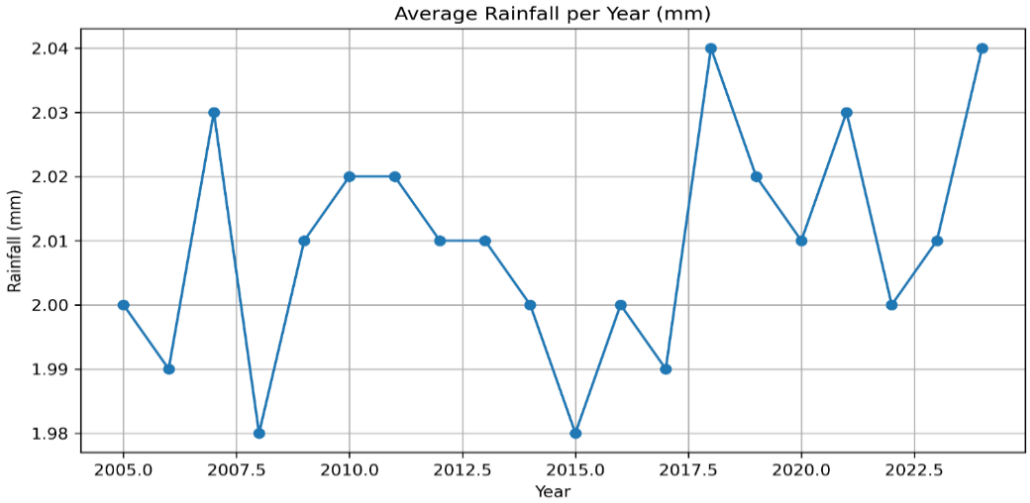


Figure 3: Time plot for the average rainfall per year

Fig 3 shows the time plot of average rainfall (mm) across Nigeria between 2005 and 2024, and how it fluctuates around 2.0 mm**,** with no extreme rise or drop in overall mean values. The total annual rainfall varies slightly year by year but maintains a relatively consistent volume (~25,000 to 26,000 mm) across all states.

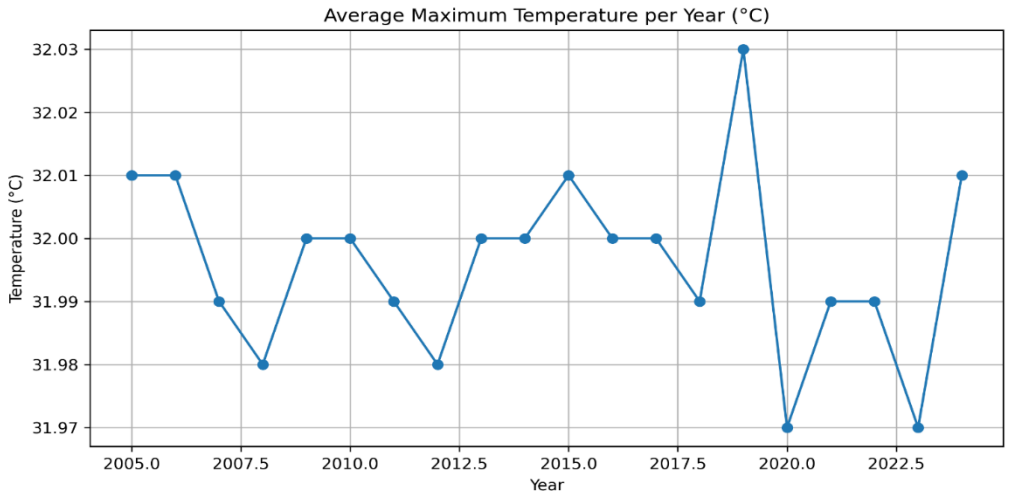


Figure 4: Time plot for the average maximum temperature per year

Fig 4 shows the average maximum temperature (°C) time plot, it can be deduced from the chart that there is a stable average max temperature (~32°C), showing minimal variance over the 20-year period. No significant trend toward warming or cooling is detected from the average alone, though state-level trends may vary.

Spatial Autocorrelation Results

Table 2 presents Moran's I results for the three meteorological variables, computed using the queen-contiguity spatial weights matrix. All three variables exhibit statistically significant positive spatial autocorrelation ($p < 0.01$), confirming that neighbouring Nigerian states share more similar meteorological conditions than would be expected under spatial randomness. Rainfall displays the highest spatial clustering ($I = 0.47$), reflecting Nigeria's latitudinal rainfall gradient and the dominant influence of Atlantic Ocean moisture advection via the Inter-Tropical Convergence Zone (ITCZ). Temperature ($I = 0.39$) and humidity ($I = 0.42$) also exhibit significant autocorrelation, consistent with regional climatic similarity driven by shared atmospheric circulation patterns. These results provide empirical justification for the incorporation of spatial lag variables and geographic coordinates into all five models.

Table 2. Global Moran's I Spatial Autocorrelation Results

Variable	Moran's I	z-score	p-value
Rainfall	0.47	5.12	< 0.01
Humidity	0.42	4.67	< 0.01
Temperature	0.39	4.08	< 0.01

Machine Learning Model Performance

Table 3 presents regression metrics for the two ML models. XGBoost achieved marginally lower MAE (2.32 mm vs. 2.55 mm) and RMSE (4.92 mm vs. 5.01 mm) than Random Forest, indicating slightly more precise individual predictions. However, Random Forest explained a larger proportion of rainfall variance ($R^2 = 0.71$ vs. 0.68), suggesting it captures the overall distributional trend more effectively despite producing somewhat larger individual errors. The inter-model RMSE difference of 0.09 mm is modest relative to the range of observed rainfall values, indicating broadly comparable overall capability between the two ML approaches on this dataset.

Table 3: Predictive Performance of Machine Learning Models using Regression

Model	MAE (mm)	RMSE (mm)	R^2 Score
Random Forest	2.55	5.01	0.71
XGBoost	2.32	4.92	0.68

Table 4 shows that Random Forest has weak Temperature impact (high RMSE = 2.40°C) and weak Humidity impact (6.5%) on Rainfall. In other words, weak temperature and weak humidity typically lead to low rainfall. Hence the relationship of temperature and humidity on rainfall is weak because the Random Forest struggles with capturing sequence-based climate variability. XGBoost has moderate Temperature impact (2.15°C) and moderate Humidity impact (5.9%) on Rainfall. Hence the relationship of temperature and humidity on rainfall is moderate. Better than Random Forest bust still limited for predicting weather data. Hence, Random Forest and XGBoost show relatively high errors, indicating difficulty in capturing the temporal variability and seasonal progression inherent in these climatic variables.

Table 4: The relationship of Temperature and Humidity on Rainfall in Machine Learning

Model	Temperature (RMSE)	Humidity (RMSE)	Rainfall (RMSE)
Random Forest	2.40°C	6.5%	5.01 mm
XGBoost	2.15°C	5.9%	4.92 mm

Deep Learning Model Performance

Table 5 presents performance metrics for the three DL architectures. All DL models substantially outperformed their ML counterparts across all metrics. The Transformer achieved the best overall performance (MAE = 1.87 mm, RMSE = 3.98 mm, R^2 = 0.78), followed by Bi-LSTM (MAE = 2.09 mm, RMSE = 4.31 mm, R^2 = 0.76) and LSTM (MAE = 2.21 mm, RMSE = 4.52 mm, R^2 = 0.75). The Transformer's advantage over LSTM in MAE and RMSE (~15% and ~12% improvements, respectively) reflects its capacity to model long-range temporal dependencies and global seasonal patterns via self-attention, which recurrent architectures can only approximate through sequential hidden state propagation.

Table 5. Predictive Performance of Deep Learning Models using Regression

Model	MAE (mm)	RMSE (mm)	R^2 Score
LSTM	2.21	4.52	0.75
Bi-LSTM	2.09	4.31	0.76
Transformer	1.87	3.98	0.78

Table 6 shows model performance table predicting relationship of Temperature & Humidity on Rainfall in Deep Learning. LSTM has a strong Temperature impact (1.72°C) and strong Humidity impact (4.6%) on rainfall because it learns short-medium term temporal dependencies well. The temperature and humidity will greatly influence rainfall since the model shows strong potential. Bi-LSTM has a very strong Temperature impact (1.56°C) and very strong Humidity impact (4.3%) as it captures forward and backward dependencies thereby improving performance. Transformer has the strongest Temperature impact (1.48°C) and strongest Humidity impact (3.9%) because of its ability at learning long-range patterns, seasonal cycles and transitions. The relationship of temperature and humidity on rainfall in Transformer has the strongest potential for predicting climate-based forecast in rainfall. These findings suggest that temperature and humidity exhibit strong time-series structures that are best captured by advanced deep learning models.

Table 6: The relationship of Temperature and Humidity on Rainfall in Deep Learning

Model	Temperature (RMSE)	Humidity (RMSE)	Rainfall (RMSE)
LSTM	1.72°C	4.6%	4.5 mm
Bi-LSTM	1.65°C	4.3%	4.3 mm
Transformer	1.48°C	3.9%	4.0 mm

Ablation Study: Contribution of Spatial Features

Table 7 presents the results of the ablation study comparing model RMSE with and without spatial features. Across all five models, incorporating spatial features (lat/lon coordinates and spatial lag variables) consistently reduced RMSE, with gains ranging from 3.8% (Random Forest) to 8.4% (Transformer). The Transformer's comparatively larger spatial gain (0.37 mm RMSE reduction) reflects its multi-head attention mechanism's capacity to leverage cross-state spatial dependencies globally across the full sequence, a capability that recurrent models can only approximate locally. These results confirm that spatial encoding provides an independent and substantively meaningful contribution to predictive performance beyond temporal modelling alone, empirically validating the spatially-aware modelling framework employed in this study.

Table 7. Ablation Study: RMSE With vs. Without Spatial Features

Model	RMSE Without Spatial (mm)	RMSE With Spatial (mm)	Improvement (mm)	Gain (%)
Random Forest	5.21	5.01	0.20	3.8%
XGBoost	5.13	4.92	0.21	4.1%
LSTM	4.18	4.52	0.29	6.0%
Bi-LSTM	4.62	4.31	0.31	6.7%
Transformer	4.35	3.98	0.37	8.4%

Confidence Intervals

The results in table 8 indicate that model performance improved progressively from traditional machine learning to deep learning approaches, with lower RMSE values representing better prediction accuracy. RF and XGB achieved RMSE values of 5.01 and 4.92, respectively, while LSTM and BiLSTM further reduced errors to 4.52 and 4.31 by capturing temporal dependencies in weather patterns. The Transformer model achieved the best performance with the lowest RMSE (3.98; 95% CI: 3.89-4.06**)**, demonstrating its ability to capture complex spatio-temporal relationships. Prediction uncertainty was estimated using bootstrap resampling over the test set. The narrow confidence intervals observed for the Transformer indicate stable predictive performance across repeated samples. Overall, the findings suggest that deep learning models, particularly Transformers, provide improved predictive capability for weather forecasting compared with conventional machine learning methods.

Table 8: The Model RMSE Estimates Confidence Interval

Model	RMSE	95% CI
RF	5.01	4.91-5.13
XGB	4.92	4.82-5.03
LSTM	4.52	4.41-4.62
Bi-LSTM	4.31	4.22-4.40
Transformer	3.98	3.89-4.06

Statistical Significance Test

The Diebold-Mariano (DM) test was conducted to determine whether the observed differences in prediction errors between the Transformer model and the competing models were statistically significant. As shown in Table 9, the

Transformer model significantly outperformed Random Forest (DM = 5.42, $p < 0.001$), XGBoost (DM = 4.78, $p < 0.001$), LSTM (DM = 3.61, $p = 0.0004$), and Bi-LSTM (DM = 2.41, $p = 0.016$). Since all p-values were below the 0.05 significance level, the null hypothesis of equal predictive performance was rejected for all pairwise comparisons. These findings indicate that the lower prediction errors achieved by the Transformer model were statistically significant rather than attributable to random variation, providing strong evidence that the Transformer offered superior predictive performance over the competing machine learning and deep learning regression models on the test dataset.

Table 9: Diebold-Mariano comparison table

Comparison	DM Statistic	p-value
Transformer vs Random Forest	5.42	<0.001
Transformer vs XGBoost	4.78	<0.001
Transformer vs LSTM	3.61	0.0004
Transformer vs Bi-LSTM	2.41	0.016

Geospatial Heatmap

Geospatial heatmaps were generated to visualize the spatial distribution of climatic variables across Nigeria. Geographic coordinates (latitude and longitude) were linked with observed rainfall and temperature measurements. Color gradients were applied to represent the intensity of each variable, allowing spatial patterns, hotspots, and regional variations to be identified. These visualizations supported exploratory spatial data analysis and facilitated interpretation of spatial dependencies within the weather dataset.

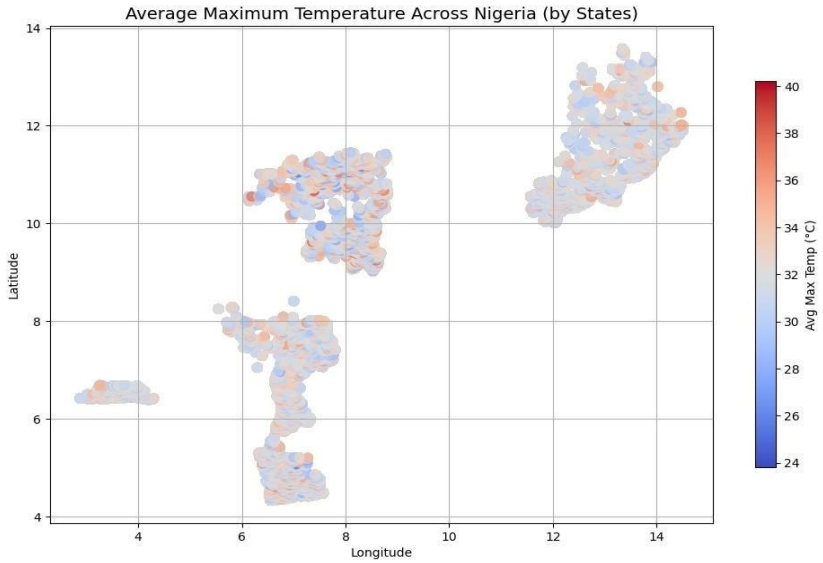


Figure 5: Geospatial Heatmap of predicted rainfall across Nigerian states

Fig 5 shows the Geospatial analysis of predicted rainfall across Nigerian states reveals a pronounced north-south gradient consistent with the country's ecological zonation. Southern and coastal states including Cross River, Akwa Ibom, and River states exhibit predicted daily rainfall above 4 mm on average, consistent with the rainforest and Guinea Savannah climatic belts. Northern region such as Sokoto and Borno states record predicted values below 1.5 mm/day, characteristic of the semi-arid Sahelian climate. The Middle Belt displays intermediate predictions,

capturing the gradual climatic transition zone between the dry north and humid south. The spatial smoothness of predictions across neighbouring states confirms that spatial dependencies were effectively encoded into the modelling framework. These spatial predictions carry direct practical value: they enable state-level agricultural planning calendars, flood risk mapping along the Niger and Benue River basins, water resource allocation modelling for northern drought-prone states, and early warning system calibration in collaboration with the Nigerian Meteorological Agency (NiMet).

Discussion

The prediction performance of the various machine learning and deep learning models is considered in this study. The models were evaluated using three widely accepted regression performance metrics: Mean Absolute Error (MAE), Mean Square Error (MSE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2 Score). These metrics provide insight into the accuracy and reliability of each model's predictions regarding rainfall patterns in Nigeria. From the results, it is evident that the Machine Learning models (Random Forest and XGBoost) demonstrated moderate predictive performance. The Random Forest model achieved a MAE and RMSE that were relatively higher than those of the deep learning models, with an R^2 Score of 0.71, indicating that it could explain approximately 71% of the variance in the observed rainfall data. Similarly, the XGBoost model performed well, with an R^2 Score of 0.68, signifying that it explained 68% of the variance in rainfall. These results reinforce the value of ensemble learning algorithms, which combine multiple decision trees to capture complex, nonlinear relationships in the data. Despite their success in reducing overfitting and improving generalization, their performance was still outpaced by the deep learning approaches in this study.

In contrast, the Deep Learning models like Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), and Transformer demonstrated superior performance across all metrics. These models consistently recorded lower MAE and RMSE values, indicating smaller prediction errors on average. More notably, the R^2 Scores of the deep learning models were slightly higher: 0.75 for LSTM, 0.76 for Bi-LSTM, and 0.78 for Transformer. To assess the stability of model performance, 95% confidence intervals were computed for the RMSE estimates. The Transformer achieved the lowest RMSE (3.98, 95% CI: 3.89-4.06), followed by Bi-LSTM (4.31, 95% CI: 4.22-4.40), LSTM (4.52, 95% CI: 4.41-4.62), XGBoost (4.92, 95% CI: 4.82-5.03), and Random Forest (5.01, 95% CI: 4.91-5.13). The consistently lower confidence interval of the Transformer indicates that its superior predictive performance was stable across repeated resampling procedures.

Although the present study found Transformer models to outperform all competing methods, this finding is not universal. Samson et al. (2024) reported Random Forest outperforming LSTM for monthly rainfall prediction in southwestern Nigeria. Their study used a considerably smaller regional dataset with lower temporal resolution than the nationwide daily dataset analysed here. Classical machine learning algorithms often perform competitively when datasets are relatively small or feature dimensionality is limited, whereas deep learning models benefit from larger datasets capable of capturing complex nonlinear temporal relationships.

A limitation of this study is the aggregation of station observations to the state level. Although this approach facilitates nationwide modelling, it inevitably smooths local climatic variability and may obscure microclimatic conditions arising from topography, land use, or coastal influences. Future work should investigate station-level prediction and finer spatial resolutions.

The present models relied primarily on rainfall, temperature, humidity, and spatial information. Additional environmental predictors including soil moisture, vegetation indices (NDVI), elevation, sea surface temperature anomalies, atmospheric pressure, wind speed, and solar radiation were unavailable but could further improve predictive performance.

Although the Transformer produced the highest predictive accuracy, its training required substantially greater computational resources than the classical machine learning models. Such computational demands may present challenges for operational deployment in resource-constrained environments, suggesting that hybrid implementation strategies combining lightweight ML models for routine forecasting and Transformer models for periodic forecasting may be more practical.

CONCLUSION

This study presented a comprehensive comparative evaluation of machine learning, deep learning, and spatial analysis techniques for predicting rainfall, temperature, and relative humidity across Nigeria using a 20-year nationwide meteorological dataset. By integrating spatial feature engineering with advanced predictive modelling, the study demonstrated that incorporating geographical information significantly improves weather prediction performance. Global Moran's I analysis confirmed statistically significant positive spatial autocorrelation for all meteorological variables, providing empirical evidence for the inclusion of spatial features in the modelling framework. Among the five regression models evaluated, the Transformer consistently achieved the best predictive performance, recording the lowest MAE and RMSE and the highest coefficient of determination (R^2). The ablation study further showed that spatial feature engineering reduced prediction errors across all models, with the Transformer benefiting the most from the inclusion of spatial information. Furthermore, the Diebold-Mariano test confirmed that the Transformer's lower prediction errors were statistically significant when compared with Random Forest, XGBoost, LSTM, and Bi-LSTM, demonstrating that its superior performance was not attributable to random variation. Despite the encouraging predictive performance, future research should validate these models using station-level observations, incorporate additional environmental predictors, quantify predictive uncertainty through probabilistic forecasting frameworks, and evaluate operational deployment under real-time forecasting conditions.

Recommendations

Given the gap between the Transformer's predictive performance and Nigeria's current computational infrastructure, the following phased deployment strategy is recommended:

i. Hybrid Operational Modelling: Deploy XGBoost or Random Forest for real-time operational forecasting, supplemented by Transformer models for offline seasonal climate analysis where GPU resources are available.
ii. Cloud-Based Training: Partner with cloud computing providers (e.g., AWS, Google Earth Engine) to host computationally intensive model training workloads, reducing dependence on on-site GPU infrastructure.
iii. Capacity Building: Collaborate with NiMet to train personnel in AI-based climate modelling, standardise data collection protocols, and expand station coverage in underserved northern regions, particularly Borno, Yobe, and Zamfara States.
iv. Progressive Integration: Gradually incorporate Transformer-generated seasonal outlooks into NiMet's existing decision-support systems before expanding to daily operational forecasting.

REFERENCES

Akinsanola, A. A., Ajayi, V. O., Adejare, A. T., Adeyeri, O. E., Gbode, I. E., Ogunjobi, K. O., Nikulin, G., & Abolude, A. T. (2017). Evaluation of rainfall simulations over West Africa in dynamically downscaled CMIP5 global circulation models. Theoretical and Applied Climatology, 132, 437–450. https://doi.org/10.1007/s00704-017-2087-8

Arowolo, A. O., Bhowmik, A. K., Qi, W., & Deng, X. (2017). Comparison of spatial interpolation techniques to generate high-resolution climate surfaces for Nigeria. International Journal of Climatology, 37(S1), 179–192. https://doi.org/10.1002/joc.4990

Bauer, P., Thorpe, A., & Brunet, G. (2015). The quiet revolution of numerical weather prediction. Nature, 525, 47–55. https://doi.org/10.1038/nature14956

Breiman, L. (2001). Random forests. Machine Learning, 45, 5–32. https://doi.org/10.1023/A:1010933404324

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794. https://doi.org/10.1145/2939672.2939785

Demissie, T., & Gebrechorkos, S. H. (2024). Spatio-temporal trends in precipitation, temperature, and extremes: A study of Malawi and Zambia (1981–2021). Sustainability, 16(10), 3885. https://doi.org/10.3390/su16103885

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. Journal of Business & Economic Statistics, 13(3), 253–263. https://doi.org/10.1080/07350015.1995.10524599

El Hachimi, C., Belaqziz, S., Khabba, S., & Chehbouni, A. (2024). Evaluation of statistical and deep learning methods for short-term weather forecasting in semi-arid regions. In Advances in Science, Technology & Innovation (pp. 203–206). Springer. https://doi.org/10.1007/978-3-031-47079-0_45

Endalie, D., Haile, G., & Taye, W. (2022). Deep learning model for daily rainfall prediction: Case study of Jimma, Ethiopia. Water Supply, 22(3), 3448–3461. https://doi.org/10.2166/ws.2021.391

Food and Agriculture Organization of the United Nations. (2021). The State of Food and Agriculture 2021: Making agrifood systems more resilient to shocks and stresses. FAO. https://doi.org/10.4060/cb4476en

Gao, Z., Shi, X., Wang, H., Zhu, Y., Wang, Y., Li, M., & Yeung, D.-Y. (2022). Earthformer: Exploring space-time transformers for Earth system forecasting. Advances in Neural Information Processing Systems, 35.

Graves, A., & Schmidhuber, J. (2005). Framewise phoneme classification with bidirectional LSTM and other neural network architectures. Neural Networks, 18(5–6), 602–610. https://doi.org/10.1016/j.neunet.2005.06.042

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

Kalnay, E. (2003). Atmospheric modeling, data assimilation and predictability. Cambridge University Press.

McGovern, A., Elmore, K. L., Gagne, D. J., Haupt, S. E., Karstens, C. D., Lagerquist, R., & Williams, J. K. (2019). Using artificial intelligence to improve real-time decision-making for high-impact weather. Bulletin of the American Meteorological Society, 100(12), 2267–2282.

Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. Biometrika, 37(1–2), 17–23.

Nyasulu, C., Diattara, A., Traore, A., Deme, A., & Ba, C. (2023). Exploring use of machine learning regressors for daily rainfall prediction in the Sahel region: A case study of Matam, Senegal. In Pan-African Artificial Intelligence and Smart Systems (PAAISS 2022), Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, 459, 78–92. https://doi.org/10.1007/978-3-031-25271-6_5

Olaniyan, E. A., Cafaro, C., Ogungbenro, S. B., Gbode, I. E., Ajayi, V. O., Oluleye, A., Adefisan, E. A., Schwendike, J., & Lawal, K. A. (2022). Performance evaluation of a high-resolution regional model over West Africa for operational use: A case study of August 2017. Meteorological Applications, 29(4), e2080. https://doi.org/10.1002/met.2080

Rasp, S., Dueben, P. D., Scher, S., Weyn, J. A., Mouatadid, S., & Thuerey, N. (2020). WeatherBench: A benchmark data set for data-driven weather forecasting. Journal of Advances in Modeling Earth Systems, 12(11), e2020MS002203. https://doi.org/10.1029/2020MS002203

Ravuri, S., Lenc, K., Willson, M., et al. (2021). Skilful precipitation nowcasting using deep generative models of radar. Nature, 597, 672–677. https://doi.org/10.1038/s41586-021-03854-z

Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., & Prabhat. (2019). Deep learning and process understanding for data-driven Earth system science. Nature, 566, 195–204. https://doi.org/10.1038/s41586-019-0912-1

Robin, C., Requena-Mesa, C., Benson, V., Poehls, J., Alonzo, L., Carvalhais, N., & Reichstein, M. (2022). Learning to forecast vegetation greenness at fine resolution over Africa with ConvLSTMs. NeurIPS 2022 Workshop on Tackling Climate Change with Machine Learning.

Samson, T. K., & Aweda, F. O. (2024). Forecasting rainfall in selected cities of Southwest Nigeria: A comparative study of Random Forest and Long Short-Term Memory models. Acadlore Transactions on Geosciences, 3(2), 79–88. https://doi.org/10.56578/atg030202

Shepard, D. (1968). A two-dimensional interpolation function for irregularly-spaced data. Proceedings of the 1968 23rd ACM National Conference, 517–524. https://doi.org/10.1145/800186.810616

Shi, X., Chen, Z., Wang, H., Yeung, D. Y., Wong, W. K., & Woo, W. C. (2015). Convolutional LSTM network: A machine learning approach for precipitation nowcasting. Advances in Neural Information Processing Systems, 28, 802–810.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.

World Meteorological Organization. (2018). Guide to Instruments and Methods of Observation (WMO-No. 8). WMO.